LESSON PLAN

ROCKET ENGINE (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Comparison of composite with conventional metal w.r.t stress-strain graphical diagram
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Comparison of composite with conventional metal w.r.t stress-strain graphical diagram.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the development of composite materials, types and functions of fibers and resin, Defects in composite, selection criteria for Matrix, fabric weaves, prepress, strength flowcharts

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of stress-strain diagrams for metal and composite materials; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between isotropic and anisotropic materials?
(b) Define the elastic limit in the context of metal stress-strain curves.
(c) List two common defects found in composite materials.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Development of composite materials • Types and functions of fibers • Functions of resin matrix • Selection criteria for matrix	L & Ds	PPT, IB	15 min
(b)	• Stress-strain diagram for conventional metal • Stress-strain diagram for composite materials • Comparison of yield and ultimate strength • Fabric weaves and prepress effects	L & Ds	PPT, PDF	20 min
(c)	• Defects in composite materials • Strength flowcharts for selection • Impact of defects on stress distribution • Summary of metal vs composite behavior	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the stress-strain curve of a composite differs from that of a metal regarding failure points.
(b) Describe the role of fabric weaves in determining the mechanical properties of a composite.

12. Home Assignment and Follow Up	Prepare a brief note comparing the failure modes of metal and composite materials as discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fatigue and notch sensitivity in composites. Factors affecting notch sensitivity. Significance of notch sensitivity during in service use. Defects in composites.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min